

EE-102 Electric Circuit Analysis

Experiment # 2

Verification of Ohm's Law

Objective/s:

- Ohm's Law verification

Apparatus

- Digital Multimeter (DMM),
- Resistors,
- Power Supply,
- Wire Stripper,
- Resistors: 100Ω, 3.6kΩ, 4.7kΩ, 6.8kΩ, 10kΩ

1. Ohm's Law

Ohm's law mathematically describes how voltage 'V', current 'I', and Resistance 'R' in a circuit are related. According to this law:

"The current in a circuit is directly proportional to the applied voltage and inversely proportional to the circuit resistance".

1.1. Formula for Voltage

For a constant value of R, V is directly proportional to I

i.e. $V = IR$

1.2. Formula for Current

For a constant value of V, I is inversely proportional to R

i.e. $I = V/R$

The way in which we accomplish this is to measure the voltage across and current through a known resistor for several different pair of values. We can then plot the data on a graph, and if the relationship is truly linear it should yield a straight line.

When graphing data such as those obtained in this experiment, either the x or y-axis can be chosen to display the voltage or current values. When the y-axis is chosen as the voltage axis, and the x-axis as the current axis, we say that we are plotting V versus I. the slope of the line

$\Delta V/\Delta I$ should be equal to the resistance R of the resistor. If on the other hand, current is plotted on the y-axis, and voltage along the x-axis, then slope of the line $\Delta I/\Delta V$ is equal to the conductance G of the resistor. In this experiment you will plot I in mA versus V in volts, and therefore the slope will be the conductance of R . When plotting a straight line on a graph such as this, it is important that you draw the best straight line that you can through the data points.

2. Power in DC Circuit

Electronic devices and circuits require energy to operate. Power is a measure in watts (W) of the energy in joules (J) consumed by a given device in one second. For a resistor, three equations will yield the power dissipated: $P=IV$, $P=V^2/R$ and $P=I^2R$. In this experiment, you will verify these formulae and plot graphs of the power versus the current, and then power versus the voltage. The resulting curves are parabolas, and the equations of the curves are called quadratic. By plotting two curves for both 1 k Ω and 390 Ω resistors on the same scales and axes, you will be able to see which resistance consumes more power for a given current (or voltage).

3. Tasks

1. Perform the following steps to verify that

$$I = \frac{V}{R}$$

- a) Measure the resistance of the 6.8k Ω (color-coded value) resistor with the DMM.

$R_{1\text{measured}} =$ _____,

- b) Connect the circuit in Figure-1.

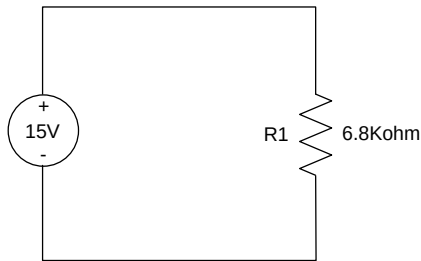


Figure 1

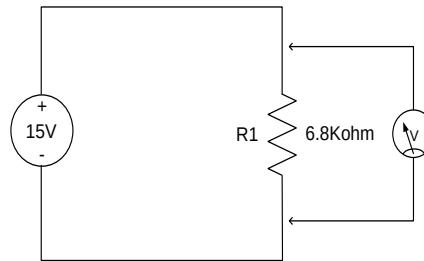


Figure 2

- c) Adjust the power supply voltage to 15.0 V using the DMM set on its DC voltage mode.
 d) Measure and record the voltage across the R_1 resistor. (As shown in Figure-2)

$V_{R1} =$ _____. Make sure that the voltage across R_1 is still 15V.

- e) Measure and record the current through R_1 . (as shown in Figure-3).

$I_{R1} =$ _____ $P = VI =$ _____

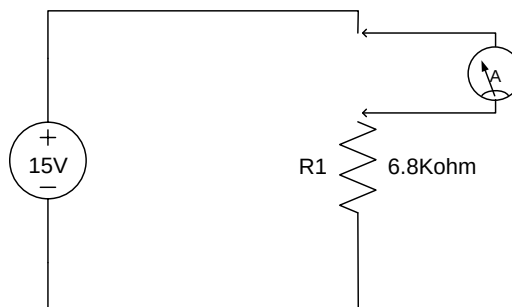


Figure 3

- f) Using the measured values of voltage and resistance, calculate the current through R_1 using Ohm's Law. Show calculations in the space below.

$$I = \frac{V}{R} = \text{_____} \quad I_{R1} = \text{_____}$$

- g) What conclusion can you make from these procedures?

2. Perform the following steps to verify that

$$R = \frac{V}{I}$$

- Connect the same circuit as in Figure 1, except change the value of resistance of 3.9k Ω (color-coded value).
- Measure and record the voltage across R_1 and the current flow through R_1 .
 $V_{R1} = \underline{\hspace{2cm}}$ $I_{R1} = \underline{\hspace{2cm}}$
- Calculate the resistance of R_1 using the measured values of V_{R1} and I_{R1} with Ohm's Law. Show calculation. $R_1 = \underline{\hspace{2cm}}$
- Remove R_1 from the circuit. Measure and record the resistance of R_1 using the multimeter.

$$R_1 = \underline{\hspace{2cm}} \quad P = I^2 R = \underline{\hspace{2cm}}$$

- What conclusion can you make from these procedures?

3. Perform the following procedures to verify that

$$V = IR$$

- Connect the circuit in Figure 1, except change the resistor to 10k Ω . (color-coded value)
- Measure the resistance of this resistor making sure that the resistor is removed from the circuit. $R_1 = \underline{\hspace{2cm}}$
- Reinsert the resistor into the circuit and measure the current through the resistor.
 $I_{R1} = \underline{\hspace{2cm}}$
- Calculate, using Ohm's law, the resistor voltage using the measured values of resistance and current. Show calculation below.

$$V = I \times R = \underline{\hspace{2cm}} \text{ mA} \times \underline{\hspace{2cm}} \Omega = \underline{\hspace{2cm}} \text{ V}$$

- Measure the resistor voltage. Record

$$V_{R1} = \underline{\hspace{2cm}} \quad P = \frac{V^2}{R} = \underline{\hspace{2cm}}$$

- What conclusion can you make from these procedures?

4. Perform the following procedures to verify that

IF THE RESISTANCE IS HELD CONSTANT, INCREASING THE VOLTAGE WILL INCREASE THE CURRENT.

- Connect the circuit in Figure 1 using a 4.7k Ω resistor.
- Set the voltage source to 2.0 V.
- Measure the current through the resistor. Record this value in the following table.
- Repeat this procedure from 2.0V to 30.0V. And note down the required values in the table.

V_{R1}	I_{R1}	P_{R1}	R_1
0V			4.7k Ω
4V			
6V			
8V			
10V			